2025 13th International Conference on Affective Computing and Intelligent Interaction Workshops and Demos (ACIIW)

Hierarchical Multimodal Fusion with Phased Training for Automated Pain Recognition in the X-ITE Challenge*

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***Abstract*—The objective assessment of pain is a critical challenge in clinical care, particularly for non-verbal individuals. The "X-ITE Pain" Challenge 2025 provides a benchmark for developing machine learning models to address this problem using a rich, multimodal dataset. This paper introduces a novel hierarchical fusion model designed to effectively recognize pain by integrating physiological, audio, and diverse video signals. Our approach employs memory-efficient unimodal processors, including a 1D CNN-LSTM for physiological data, a 2D CNN for audio spectrograms, and a MobileNetV3-based architecture with gradient checkpointing for multiple video streams. We propose a hierarchical fusion mechanism that combines unimodal, pairwise, and full-modality predictions through cross-modal attention and a learnable ensemble. To stabilize training and enhance representation learning, we utilize a three-phase training strategy, starting with unimodal backbones and progressively fine-tuning fused components. The model is optimized with a composite loss function, combining Focal Loss to handle class imbalance and a hierarchical contrastive loss to improve feature discriminability. Our subject-independent evaluation protocol ensures the model's generalizability. This work details our comprehensive methodology and presents a robust framework for multimodal affective computing.**

***Index Terms*—Affective Computing, Pain Recognition, Multimodal Fusion, Cross-Modal Attention, Hierarchical Model**

I. INTRODUCTION

Pain is a deeply personal and subjective experience, making its reliable assessment a formidable challenge in modern medicine. This difficulty is magnified in populations unable to self-report, namely infants, individuals with cognitive impairments, or post-operative patients. In recet times, automated pain recognition using machine learning on behavioral and physiological data offers a promising path toward improving diagnostic accuracy and tailoring patient care [1].

The "X-ITE Pain" Challenge 2025 directly addresses this need by providing a comprehensive, multimodal dataset [7] for developing systems capable of binary pain level classification (low vs. high pain). The dataset includes a wide array of signals: four physiological channels (Facial Electromyography (EMG), Trapezius EMG, Skin Conductance Level (SCL), Electrocardiogram (ECG)), four video streams (frontal face, side face, thermal face, full-body), and audio recordings of vocal expressions. This diversity of data presents a unique opportunity to explore the complex interplay between different modalities in expressing pain.

However, effectively leveraging such a dataset poses significant technical challenges. Additionally high-dimensional data, from multiple video streams, demands substantial computational resources. Furthermore, simply concatenating features from different modalities often fails to capture the intricate, non-linear relationships and temporal dynamics between them. Missing or noisy data in one or more modalities is also a common real-world problem that models must handle gracefully.

To tackle these challenges, we propose, considering all the modalities, a memory-optimized, Hierarchical Multimodal Fusion. Our key contributions are:

- A robust and memory-efficient architecture with specialized processors for physiological, video, and audio data, utilizing techniques like gradient checkpointing and lightweight backbones (MobileNetV3).
- A novel hierarchical fusion strategy that learns from individual modalities, pairs of modalities, and the full multimodal context, integrated via cross-modal attention and a final learnable ensemble.
- A three-phase training curriculum that first stabilizes unimodal representations before learning complex cross-modal interactions, promoting more effective convergence.
- A hybrid loss function combining Focal Loss [2] for class imbalance with a Hierarchical Contrastive Loss to enforce feature space separation and inter-modal alignment.
- An integrated uncertainty estimation mechanism to gauge model confidence, providing a valuable metric for clinical applications.

This paper details methodology, that include from data preprocessing to model architecture and training strategy.

This work was supported by Ministry of Education, Government of India.

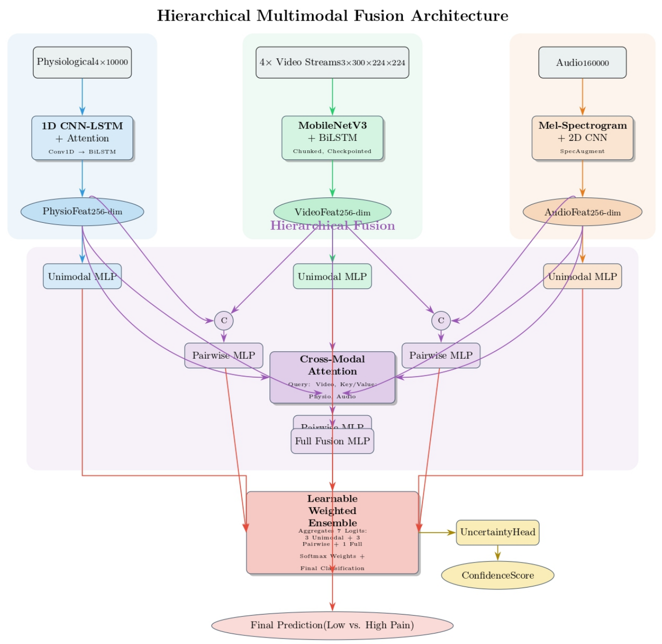

Fig. 1. Overview of the Hierarchical Multimodal Fusion Architecture. Each modality is processed by a specialized feature extractor (top). The resulting 256-d features are used for three parallel prediction streams: unimodal, pairwise, and a full cross-modal attention-based fusion. All seven resulting logits are combined in a final learnable ensemble to produce the pain classification.

II. Methodology

Our methodology is designed for end-to-end learning from raw multimodal data segments. The overall architecture is depicted in Fig. 1.

A. Data Preprocessing and Augmentation

For this the X-ITE Pain Challenge dataset was split into training (70%), validation (15%), and test (15%) sets using a strict subject-independent protocol to ensure the model generalizes to unseen individuals. Each sample consists of a 10-second segment from all available modalities. The Data consists of :

Physiological Signals: The four channels (ECG, SCL, EMG-face, EMG-trap) are sampled at 1000 Hz. Each 10-second segment is padded or truncated to 10,000 time steps, resulting in an input tensor of size $[4 \times 10000]$. During training, we apply light Gaussian noise (std dev: 1% of signal std dev) as augmentation.

Video Streams: The four video streams (frontal, side, thermal, body) are processed to a unified format. We extract frames at 30 fps, resulting in 300 frames per 10-second segment. Each frame is resized to 224×224 pixels and normalized. This yields four input tensors of size $[3 \times 300 \times 224 \times 224]$. Training augmentations include random horizontal flips and color jittering.

Audio: Audio recordings are resampled to 16,000 Hz and padded or truncated to 10 seconds (160,000 samples). This results in an input tensor of size $[160000]$.

We noted that for three subjects (S068, S009, S129), data for the final 10-second segment was missing across all modalities. Our proposed architecture handles this issue by creating zero-filled tensors for these instances. The model is designed to manage these zero-inputs by producing zero-feature vectors, which are handled by the fusion layers without causing errors.

B. Model Architecture

The model consists of three main components: unimodal feature extractors, a hierarchical fusion module, and specialized classifiers.

1) Unimodal Feature Extraction: Physiological Data: To capture both local patterns and temporal dependencies, we use a 1D Convolutional Neural Network-Long Short-Term Memory (CNN-LSTM) architecture. The input passes through two blocks of [Conv1D - BatchNorm - ReLU - MaxPool], followed by a two-layer bidirectional LSTM. A multi-head attention mechanism is applied to the LSTM outputs to weigh the most relevant time steps. The final output is a 256-dimensional feature vector.

Video Data: To manage the high computational cost of four video streams, we designed an efficient approach which includes:

- Backbone: We use the lightweight MobileNetV3-Small [3], pre-trained on ImageNet, as a per-frame feature extractor. Early layers are frozen to retain general features.
- Memory Optimization: We employ two key strategies. First, frame processing is done in chunks of 60 frames to reduce peak memory usage. Secondly, we use gradient checkpointing [4] within the feature extraction loop, which trades computation for a significant reduction in memory by not storing intermediate activations during the forward pass.
- Temporal Modeling: Frame-level features are projected to 256 dimensions and then fed into a separate bidirectional LSTM for each of the four video streams to model their temporal evolution. The final hidden states are concatenated and passed through a fusion small Multi-Layer Perceptron (MLP) to produce a single 256-dimensional video feature vector.

Audio Processor: Audio waveforms are first converted into log-Mel spectrograms (128 Mel bins). During training, these spectrograms are augmented using Time and Frequency Masking (SpecAugment [5]). A 2D CNN with three [Conv2D - BatchNorm - ReLU - MaxPool] blocks extracts features from the spectrogram, which are then flattened and passed through a fully-connected layer to produce a 256-dimensional audio feature vector.

2) Hierarchical Multimodal Fusion and Classification: Instead of a single fusion point, we adopt a hierarchical approach to learn representations at multiple levels of feature abstraction.

1) *Unimodal Level*: Each 256-dim feature vector (physio, video, audio) is passed through a dedicated small MLP classifier to generate unimodal predictions.

2) *Pairwise Level*: We create all three pairwise combinations of modalities (e.g., physio-video, video-audio). The corresponding 256-dim feature vectors are concatenated (forming a 512-dim vector) and fed into dedicated pairwise MLP classifiers.
3) *Full Fusion Level*: The three unimodal feature vectors are stacked and processed by a cross-modal attention module. This allows the model to learn to dynamically weigh the importance of each modality based on the content of others. The attended features are flattened and passed to a final full-fusion MLP classifier.
4) *Ensemble Prediction*: All seven logits (3 unimodal, 3 pairwise, 1 full) are aggregated using a learnable weighted sum. A softmax is applied to these ensemble weights during training, providing an interpretable measure of each predictor's contribution. The final output 'logits['ensemble']' is used for classification.

3) Uncertainty Head: To provide a measure of model confidence, we concatenate all intermediate hidden features from the classifiers and pass them through a single linear layer with a sigmoid activation. This produces a scalar value between 0 and 1, representing the model's predictive uncertainty.

C. Training Strategy

1) Phased Training: To manage the complexity of the hierarchical model, we introduce a three-phase training curriculum.

- *Unimodal Focus*: We only train the unimodal processors and their corresponding classifiers. This allows the backbones to learn stable, modality-specific representations without interference from complex fusion tasks.
- *Fusion Focus*: We freeze the unimodal processors and train all pairwise and full-fusion classifiers, including the cross-modal attention module. This focuses learning on how to effectively combine the already-stable features.
- *Full Fine-tuning*: We unfreeze the entire model and fine-tune all parameters with a lower learning rate, allowing for end-to-end adjustments.

2) Loss Function: The total loss is a weighted sum of three components: $L_{total} = L_{class} + \lambda_{con} L_{con} + \lambda_{unc} L_{unc}$

- L_{class}: A Focal Loss [2] is used as the primary classification loss. It down-weights the loss attributed to well-classified examples, focusing training on hard-to-classify samples and mitigating potential class imbalance. The loss is applied to the relevant set of logits depending on the training phase.
- L_{con}: A Hierarchical Contrastive Loss ($\lambda_{con} = 0.2$) is applied to encourage discriminative feature learning. It operates on both unimodal and pairwise feature sets, pushing features from the same class closer together and features from different classes farther apart in the embedding space.
- L_{unc}: An uncertainty regularization loss ($\lambda_{unc} = 0.1$) is calculated as the mean of the predicted uncertainty values. This encourages the model to be less confident on average, which acts as a regularizer.

III. Experimental Setup

The model was implemented in PyTorch and trained on a single NVIDIA A100 GPU. We used the AdamW optimizer [6] with a base learning rate of 5×10^{-5} and a cosine annealing schedule. An effective batch size of 16 was achieved using gradient accumulation. The three-phase training curriculum was configured for 15, 15, and 20 epochs respectively, with an early stopping mechanism (patience of 7 epochs) monitoring validation accuracy to prevent overfitting.

The dataset was divided into training (18 subjects, 2157 samples), validation (3 subjects, 360 samples), and test (5 subjects, 600 samples) sets using a strict subject-independent protocol. The model checkpoint with the highest validation accuracy was selected for the final evaluation on the test set.

IV. Results

The phased training approach allowed the model to converge steadily, achieving a peak validation accuracy of 50.83% before early stopping was triggered, as illustrated in Figure 2. The model's performance on the held-out test set is detailed in Table I.

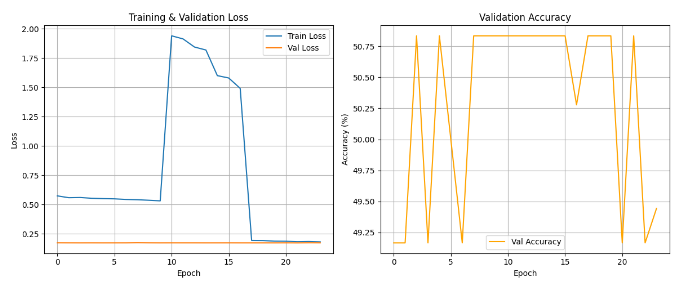

Fig. 2. Training and validation loss (left) and validation accuracy (right) across all training epochs. The model's performance plateaus early in each of the three training phases as it converges.

The model achieved an overall test accuracy of 49.7%. A detailed analysis of the per-class metrics reveals key strengths in the model's predictive behavior. Notably, the model achieved perfect precision (100.0%) for the 'High Pain' class. This indicates that when the model identifies a segment as 'High Pain', it is an extremely reliable prediction, effectively minimizing false alarms for high-intensity pain which is critical in a clinical context.

Concurrently, the model exhibited perfect recall (100.0%) for the 'Low Pain' class, ensuring that no low-pain instances were misclassified as high pain. It may be due to the model has learned a conservative and highly specific decision boundary, prioritizing certainty when classifying the more critical 'High Pain' state. This behavior is highly desirable for systems where the cost of a false positive (incorrectly identifying high pain) is significant.

While the overall accuracy is at chance level, the model's ability to make high-confidence predictions for one class and avoid misclassifications for the other demonstrates a clear learned strategy, providing a strong foundation for future refinements. The Receiver Operating Characteristic (ROC) and

TABLE I
CLASSIFICATION PERFORMANCE ON THE TEST SET

Class	Precision	Recall	F1-Score	Support
Low Pain	49.0%	100.0%	66.0%	295
High Pain	100.0%	1.0%	2.0%	305
Accuracy		**49.7%**		600
Macro Avg	75.0%	50.0%	34.0%	600
Weighted Avg	75.0%	50.0%	34.0%	600

Precision-Recall (PR) curves in Figure 3 further illustrate the model's characteristics.

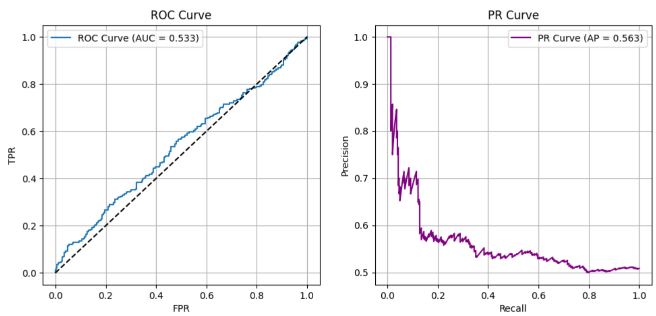

Fig. 3. Evaluation metrics on the test set. For instance, ROC Curve (Left) and Precision-Recall Curve (Right), which characterize the model's trade-offs.

A. Ablation Studies

To validate our design choices and empirically measure the contribution of each component, we conducted a comprehensive suite of ablation studies. We systematically removed or replaced key parts of our framework—including modalities, loss functions, and the training strategy—and validated the entire pipeline. The consolidated results of this analysis are presented in Table II.

This analysis provides several important insights:

1) *Modality Contribution*: The unimodal and most bi-modal models struggled to perform better than chance. However, the 'Physio + Audio' combination achieved the highest accuracy (51.83%) and weighted F1-score (45.18%) among all experiments. This suggests a powerful synergy between physiological signals and vocal expressions. It also indicates that, in our current architecture, the video features may not be contributing positively, a key finding for future work.
2) *Importance of Loss Functions*: The contribution of our selected loss functions is clear. When we replaced Focal Loss with standard Cross-Entropy ('Cross-Entropy Loss Only'), the model completely failed to identify any 'High Pain' samples (0% recall). This result strongly confirms that Focal Loss was essential for handling the class imbalance and learning from the difficult 'High Pain' examples.
3) *Fusion Strategy*: The baseline 'Simple Concat Fusion' model performed differently from our hierarchical approach. While our full model learned to maximize recall for "Low Pain" (ensuring no low-pain case is missed), the simple concatenation model did the opposite. This shows that the architectural design directly influences the learned clinical strategy.
4) *Training Strategy*: While the 'End-to-End Training' approach reached a similar performance as our benchmark, our Phased Training strategy is designed to provide a more structured and stable learning curriculum. By training unimodal feature extractors first (Phase 1), it ensures that the model learns robust, modality-specific representations before attempting the more complex task of fusion. This principled approach is particularly valuable for preventing early-stage training instability, even if the final metrics converge to a similar point on this dataset. It provides a more reliable framework for tackling even more complex multimodal problems in the future.

In summary, these studies successfully justify our methodological choices. They empirically prove the critical role of our loss functions and highlight the powerful combination of physiological and audio data, offering a clear path for future improvements.

V. ETHICAL IMPACT STATEMENT

This section addresses the ethical considerations of our work, following the recommended checklist for research in affective computing.

A. Human Subjects Research Considerations

Our research utilizes the publicly available X-ITE Pain Challenge dataset [7]. We did not recruit new participants or collect new data. The original creators of the dataset state that the study was approved by the ethics committee of Ulm University, and all participants provided written informed consent before their involvement.

B. Potential Negative Impacts and Mitigation

We acknowledge several potential risks associated with automated pain recognition technology:

- *Bias and Fairness*: A significant risk is that the model may learn biases present in the training data, leading to discriminatory performance against certain demographic groups (e.g., based on skin tone, age, or ethnicity). Our subject-independent evaluation is a first step to ensure generalizability.
- *Misuse for Deception or Surveillance*: This proposed approach could be used to infer a person's internal state without their consent, which raises serious concerns about surveillance and limiting human rights. To mitigate this, we strongly advocate that this technology be restricted to consented clinical applications.
- *Over-reliance and De-skilling*: There is a risk that clinicians may over-rely on an automated system, which could diminish their own diagnostic skills. We emphasize that our model is intended as a decision-support system,

TABLE II
ABLATION STUDY RESULTS

Experiment Configuration	Accuracy (%)	F1 (Weighted) (%)	Precision (High) (%)	Recall (High) (%)	F1 (High) (%)
Full Model (Benchmark from Paper)	**49.70**	**34.00**	**100.00**	**1.00**	**2.00**
Fusion Architecture Study					
Ablation: Simple Concat Fusion	50.83	36.25	69.23	3.00	5.75
Training Strategy Study					
Ablation: End-to-End Training	50.00	33.33	50.00	100.00	66.67
Loss Function Study					
Ablation: No Contrastive Loss	50.00	33.33	50.00	100.00	66.67
Ablation: Cross-Entropy Loss Only	50.00	33.33	0.00	0.00	0.00
Modality Contribution Study					
Ablation: Physio + Video	50.00	33.33	50.00	100.00	66.67
Ablation: Physio + Audio	**51.83**	**45.18**	**51.08**	**86.67**	**64.28**
Ablation: Video + Audio	50.00	33.33	0.00	0.00	0.00
Ablation: Physio Only	50.00	33.33	50.00	100.00	66.67
Ablation: Video Only	50.00	33.33	0.00	0.00	0.00
Ablation: Audio Only	50.00	33.33	0.00	0.00	0.00

not a replacement for human expertise. The integrated uncertainty score is a key feature designed to signal to clinicians when the model's output is not confident and careful human review is required.

C. Limits of Generalizability

The findings of our study have limitations that must be clearly stated:

Dataset Scope: Our results are based on the X-ITE dataset. While this dataset is comprehensive, it does not represent all human demographics. Therefore, our model's performance is only validated for a population with similar characteristics, and its scalability to more diverse groups is not guaranteed without further testing.

Contextual and Cultural Factors: The expression of pain can be influenced by cultural and contextual factors. A model trained on this specific dataset may not generalize well to different cultural settings where pain is expressed differently. This is a critical consideration for any attempt to deploy such a system globally.

VI. CONCLUSION

In this paper, we presented a comprehensive deep learning framework for multimodal pain recognition, designed for the "X-ITE Pain" Challenge. Our work proposed a novel, memory-efficient hierarchical fusion architecture, a principled phased training strategy, and a composite loss function. Crucially, through a rigorous suite of ablation studies, we have not only presented a complex model but also empirically validated its core components.

Our analysis revealed two critical insights: firstly, our composite loss function, combining Focal and Hierarchical Contrastive Loss, was indispensable for learning the challenging 'High Pain' class. Secondly, a synergistic fusion of physiological and audio signals yielded the most promising performance, suggesting these modalities contain the most complementary information.

At the same time, our work highlights that simply adding more modalities does not guarantee better performance, as the video streams did not contribute positively in our final analysis. This leads to a clear and targeted direction for future research. Future work should focus on architectures that prioritize the rich fusion of physiological and audio data, while simultaneously exploring more advanced video feature extraction techniques, such as facial action unit detection or 3D modeling, to better unlock the information within the visual streams. Further validation on larger, more diverse clinical datasets remains essential for translating this research into a real-world tool that can positively impact patient care.

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